MILITARY SPECIFICATION SHEET

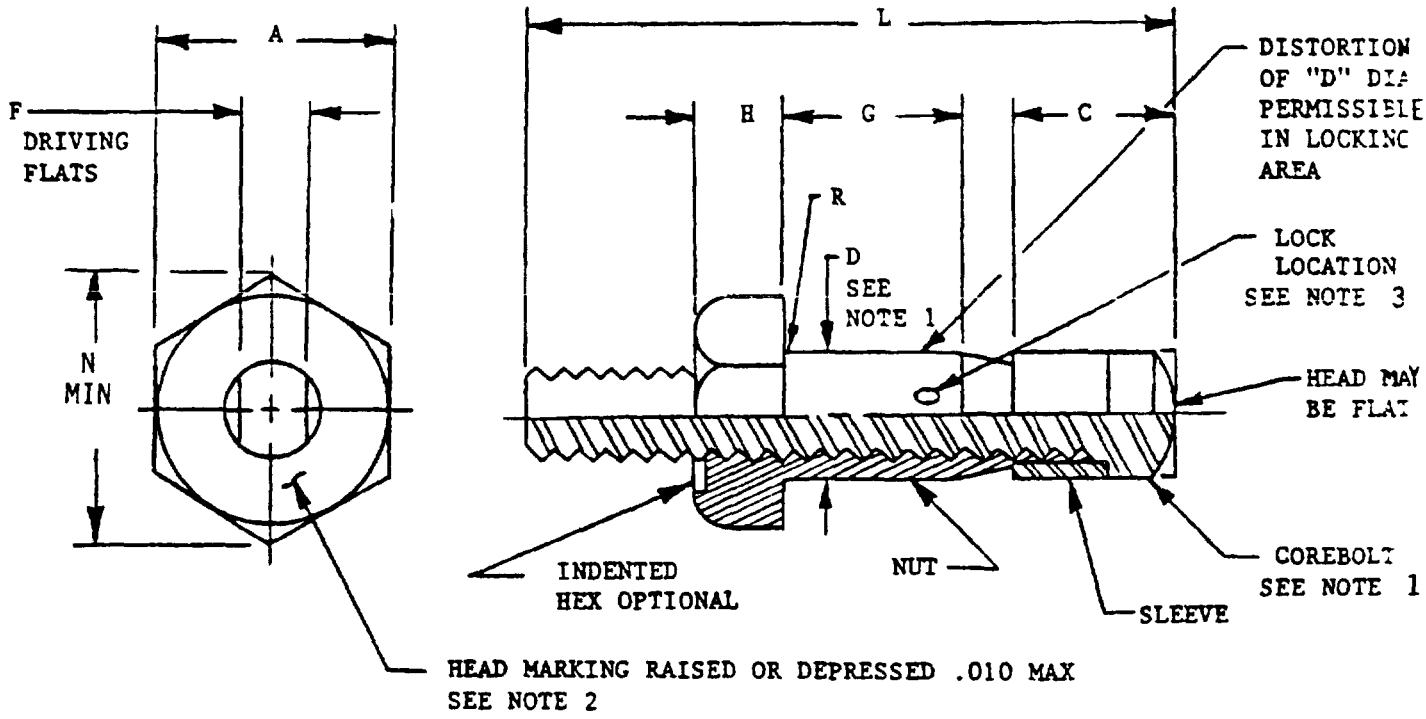
FASTENERS, BLIND, HIGH STRENGTH, INSTALLATION FORMED, THREADED, SELF-LOCKING, PROTRUDING HEAD, 450°F, ALLOY STEEL, 112 KSI, Fsu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the fasteners described herein shall consist of this document and the latest issue of specification MIL-F-81177.

Design and usage limitations - These blind fasteners shall be used in accordance with the limitations of MIL-F-81177/1.

Tooling for Installation per MIL-F-81177 - Appendix B.



NOTES:

1. Outside diameter of head of corebolt and outside diameter of sleeve shall be not greater than maximum "D" diameter.
2. Head marking shall consist of the manufacturer's identification and the numerals "250."
3. Locking feature shall be visible in this area.

FIGURE 1. Blind fastener.

FSC 5320

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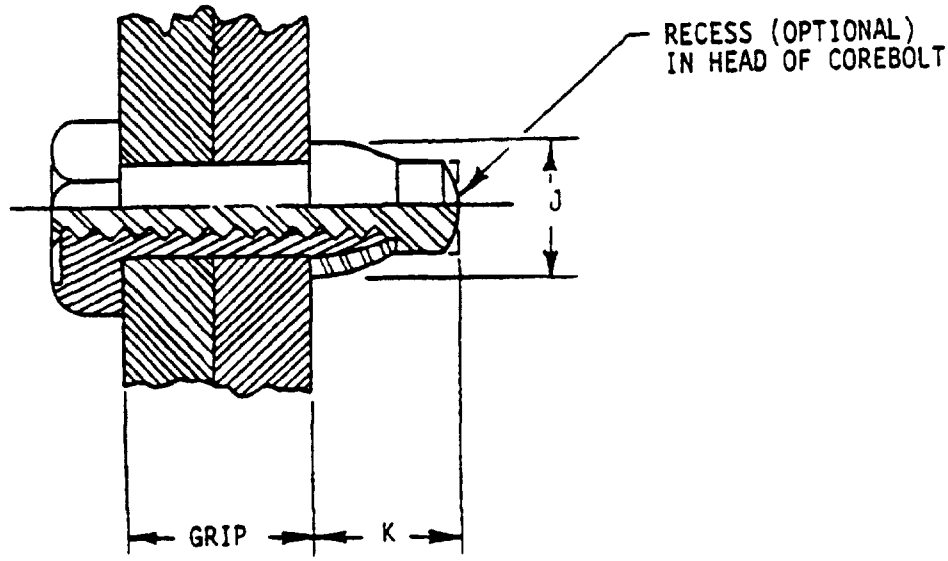


FIGURE 2. Typical installation.

REQUIREMENTS

Material:	Nut & Corebolt:	5% Chrome Steel (H-11) Per AMS 6487
	Sleeve:	Inconel, X-750 per AMS 5667 or AMS 5582, or A-286 per AMS 5732, AMS 5737, or AMS 5731
Heat Treat:	Nut & Corebolt.	Rockwell "C" 48-53
Protective treatment:	Nut & Corebolt:	Cadmium Plate per NAS 672 with Supplementary Chromate treatment per QQ-P-416, Color nut black.
	Sleeve:	Cadmium plate per QQ-P-416 Type II, Class 2.
Lubricant:	Lubricant listed below may be applied to any or all components if required for performance.	
	Cetyl Alcohol per MIL-L-87132, Lauric Acid, WWM Carbo-Wax, Dry Film Lube MIL-L-8937	

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TABLE I. Sizes and dimensions.

DASH NO.	NOM DIA.	A	C MAX	D DIA	F DRIVING FLATS	H MAX	J DIA MAX	K MAX	N MIN	R RAD	DOUBLE SHEAR STRENGTH LBS. MIN.	TENSILE STRENGTH LBS. MIN.
-5	.1635	244 .250	.287	.1645 .1625	.086 .081	.096	.244	.257	.272	.025 .020	4,680	1,350
-6	.1980	.312 305	.326	.1990 .1970	.104 .099	.113	.300	.288	.340	.030 .020	6,900	2,100
-8	.2590	.375 .429	.360	.2600 .2580	.135 .130	.135	.384	.320	.409	.035 .030	11,800	3,650
-10	.3105	.437 .429	.440	.3115 3095	.152 .147	.160	.427	.396	.479	.045 035	17,000	5,200
-12	.3735	.500 .491	.539	.3745 .3725	.185 .180	.190	.516	.473	.548	.045 .035	24,500	7,500

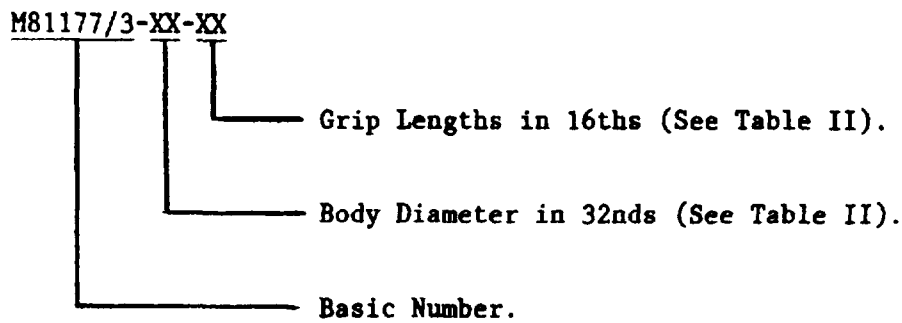
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TABLE II. Body diameters and grip lengths.

SECOND DASH NUMBER	GRIP RANGE		G ±.020 -.000	L (REF)				
	MIN	MAX		-5	-6	-8	-10	-12
-1	.031	.093	.093	.713	.850			
-2	.094	.156	.156	.775	.912	972	1.189	
-3	.157	.219	.219	.838	.975	1.035	1.252	1.324
-4	.220	.281	.281	.900	1.037	1.097	1.314	1.386
-5	.282	.344	.344	.963	1.100	1.160	1.377	1.449
-6	.345	.406	406	1.025	1.162	1.222	1.439	1.511
-7	.407	.469	.469	1.088	1.225	1.285	1.502	1.574
-8	.470	.531	.531	1.150	1.287	1.347	1.564	1.636
-9	.532	.594	.594	1.213	1.350	1.410	1.627	1.699
-10	.595	656	.656	1.275	1.412	1.472	1.689	1.761
-11	.657	.719	.719	1.338	1.475	1.535	1.752	1.824
-12	.720	.781	.781	1.401	1.537	1.597	1.814	1.885
-13	.782	.844	.844	1.463	1.600	1.660	1.877	1.949
-14	.845	.906	.906	1.526	1.662	1.722	1.939	2.011
-15	.907	.969	.969	1.588	1.725	1.785	2.002	2.074
-16	.970	1.031	1.031	1.650	1.787	1.847	2.064	2.136
-17	1.032	1.094	1.094		1.850	1.910	2.127	2.199
-18	1.095	1.156	1.156		1.912	1.972	2.189	2.261
-19	1.157	1.219	1.219		1.975	2.035	2.252	2.324
-20	1.220	1.281	1.281		2.037	2.097	2.314	2.386
-21	1.282	1 344	1.344		2.100	2.160	2.377	2.449
-22	1.345	1.406	1.406		2.162	2.222	2.439	2.511
-23	1.407	1.469	1.469		2.225	2.285	2.502	2.574
-24	1.470	1.531	1.531		2.287	2.347	2.564	2.636
-25	1.532	1.594	1.594		2.350	2.410	2.627	2.699
-26	1.595	1.656	1.656		2.412	2.472	2.689	2.761
-27	1.657	1.719	1.719		2.475	2.535	2.752	2.824
-28	1.720	1.781	1.781		2.537	2.597	2.814	2.886
-29	1.782	1.844	1.844		2.600	2.660	2.877	2.949
-30	1.845	1.906	1.906		2.662	2.772	2.989	3.011
-31	1.907	1.969	1.969		2.725	2.785	3.002	3.074
-32	1.970	2.031	2.031		2.787	2.847	3.064	3.136

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EXAMPLE OF PART NUMBER



Custodians
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Air Force - 11

Review Activities
Air Force - 99
DLA - IS
Army - AR, MI

Preparing Activity
Navy - AS

Project No. 5320-0308-3

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-81177/3 Fasteners, Blind, High STrength, Installation Formed, Threaded

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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